

# EM 613 - Mathematical Models for Data Science

## Data Pre-processing and Feature Engineering



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# Outline

- 1 Problem Description
- 2 Pre-processing Continuous Data
  - Handling missing values
  - Outlier Detection and Treatment
  - Scaling and Normalization
  - Transformation
- 3 Pre-processing Discrete and Categorical Data
  - Encoding
  - Binning
- 4 Conclusions

# Learning Outcomes

**After this lecture, students are expected to:**

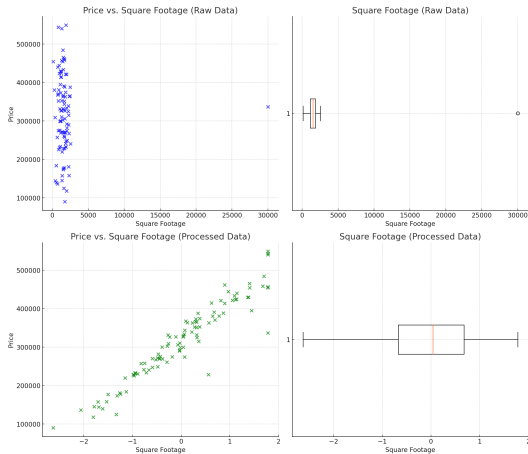
- Appreciate why pre-processing is needed in various data science problems.
- Understand various pre-processing techniques for different types of data.
- Write codes in Python to implement various pre-processing techniques discussed and improvise on the techniques not discussed.

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# Problem Description

- **Objective:** Predict the housing price (\$, INR) based on certain input features.
- **Input Features:**
  - Square Footage (sq ft)
  - Number of Bedrooms (count)
  - Neighborhood (categorical: A, B, C)
  - Age of House (years)
- **Output:**
  - Housing Price (\$, INR)

# Comparison of Raw and Processed Data



- **Left:** scatter plot and box plot on the left: Notable issues include missing values in square footage, extreme outliers (e.g., an unusually large house with 30,000 sq ft), and skewed distributions.
- **Right:** scatter plot and box plot on the right display the data after pre-processing. Missing square footage values were imputed with the mean, outliers were capped at the 95th percentile, and numerical features were standardized.

# One Size Does Not Fit All

- **Continuous Data:**
  - Examples: House prices, square footage.
  - Mathematical Approaches:
    - **Scaling:** Min-Max Scaling, Standardization.
    - **Outlier Handling:** Z-score normalization, log transformation.
- **Discrete Data:**
  - Examples: Number of bedrooms, count of items.
  - Mathematical Approaches:
    - **Normalization:** Scaling discrete values to a specific range.
    - **Binning:** Converting numerical values into categorical bins.

# One Size Does Not Fit All (cont.)

- **Categorical Data:**
  - Examples: Type of house, color.
  - Mathematical Approaches:
    - **One-Hot Encoding:** Converting categories into binary vectors.
    - **Label Encoding:** Assigning integer values to categories.
- **Ordinal Data:**
  - Examples: Customer satisfaction ratings, education level.
  - Mathematical Approaches:
    - **Ordinal Encoding:** Assigning ordered integer values to categories.
    - **Normalization:** Scaling the ordinal values.
- **Time Series Data:**
  - Examples: Stock prices, weather data over time.
  - Mathematical Approaches:
    - **Differencing:** Removing trends and seasonality.
    - **Rolling Statistics:** Applying moving averages.



# Dataset Overview

| Square Footage | Bedroom | NH | House Condition | Age of House | House Price |
|----------------|---------|----|-----------------|--------------|-------------|
| 2029.22        | 1       | B  | Poor            | 31           | 395557.69   |
| 1620.05        | 4       | A  | Poor            | 17           | 353079.59   |
| 1793.62        | 3       | B  | Fair            | 28           | 353727.78   |
| 2172.27        | 2       | C  | Fair            | 48           | 463852.14   |
| 2060.27        | 3       | A  | Poor            | 48           | 404121.51   |

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## Handling missing values

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# Handling Missing Values in Continuous Data

- **Mean/Median Imputation:**
  - Replace missing values with the mean or median of the column.
- **Interpolation:**
  - Estimate missing values using linear or polynomial interpolation.

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# Z-score Method

- Calculate z-scores for each data point.
- Identify outliers as data points with z-scores beyond a threshold (e.g.,  $\geq 3$  or  $\leq -3$ ).

# IQR Method

- Calculate the interquartile range (IQR).
- Identify outliers as data points outside  $1.5 \times \text{IQR}$  above the third quartile and below the first quartile.

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# Min-Max Scaling

- Rescale data to a fixed range, usually  $[0, 1]$ .
- Formula:  $x' = \frac{x - \min(x)}{\max(x) - \min(x)}$

# Standardization

- Rescale data to have a mean of 0 and a standard deviation of 1.
- Formula:  $z = \frac{x - \mu}{\sigma}$

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# Log Transformation

- Apply a logarithmic transformation to reduce skewness.
- Formula:  $x' = \log(x)$

# When to Use Log Transformation

## Appropriate Scenarios:

- Data with positive skewness.
- Data with heteroscedasticity (non-constant variance).
- Data spanning several orders of magnitude.

## Examples:

- Financial data (e.g., income, stock prices).
- Biological data (e.g., growth rates, population sizes).

# Box-Cox Transformation

## Definition:

- Box-Cox transformation is a family of power transformations that aim to stabilize variance and make the data more normally distributed.

## Mathematical Formula:

$$y(\lambda) = \begin{cases} \frac{y^\lambda - 1}{\lambda} & \text{if } \lambda \neq 0 \\ \log(y) & \text{if } \lambda = 0 \end{cases} \quad (1)$$

# When to Use Box-Cox Transformation Over Log Transformation

## Flexibility in Transformation:

- Box-Cox transformation can handle a variety of transformations, not just logarithmic.
- The optimal  $\lambda$  can be any real number, allowing for more tailored transformations.

# Positive Values Only

## Requirements:

- Both Box-Cox and log transformations require positive data.
- Box-Cox can be adjusted for different ranges and scales.



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# Pre-processing Discrete and Categorical Data

## Example:

- Consider a dataset of student grades in different subjects.
- Grades are categorical values: A, B, C, D, and F.
- We need to preprocess these grades for data analysis and feature engineering.

**Encoding**

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# Encoding Discrete and Categorical Data

## Why Encoding is Necessary:

- Data analysis often requires numerical input.
- Categories need to be converted into numerical values for statistical analysis.

## Common Encoding Techniques:

- Label Encoding
- One-Hot Encoding
- Ordinal Encoding

# Label Encoding

## Definition:

- Assigns a unique integer to each category.

## Example:

- Grades: A, B, C, D, F
- Encoded: 0, 1, 2, 3, 4

## Usage in Data Analysis:

- Useful for statistical models that require numerical input.
- Enables correlation analysis between categorical variables and numerical outcomes.

# One-Hot Encoding

## Definition:

- Converts each category into a new binary feature.
- Each feature is 1 if the category is present, otherwise 0.

## Example:

- Grades: A, B, C, D, F
- Encoded:  $[1, 0, 0, 0, 0]$ ,  $[0, 1, 0, 0, 0]$ ,  $[0, 0, 1, 0, 0]$ ,  $[0, 0, 0, 1, 0]$ ,  $[0, 0, 0, 0, 1]$

## Usage in Data Analysis:

- Useful for visualizing relationships between categories and numerical outcomes.
- Facilitates the use of statistical models that handle binary variables.

# Ordinal Encoding

## Definition:

- Assigns an integer to each category based on their order.

## Example:

- Grades: A, B, C, D, F
- Encoded: 4, 3, 2, 1, 0

## Usage in Data Analysis:

- Maintains the ordinal relationship between categories.
- Enables regression analysis where the order of categories matters.

**Binning**

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# Binning as a Pre-processing Technique

## Types of Binning:

- Equal-width binning: Divides the range into intervals of equal size.
- Equal-frequency binning: Each bin has the same number of observations.
- Custom binning: Bins defined based on domain knowledge or specific criteria.

## Usefulness:

- Reduces the effects of minor observation errors.
- Handles outliers effectively.
- Simplifies models by reducing the number of distinct values.

## Binning Followed by Encoding:

- Binning can be followed by encoding to convert bins into numerical values.
- Useful for models that require numerical input.

# Choosing the Right Encoding Technique

## Considerations:

- Nature of the Data: Ordinal vs. Nominal
- Impact on Analysis: How the encoding affects the interpretation of results.
- Data Size: One-Hot encoding can lead to a large number of features.

## Example:

- Grades (Ordinal): Ordinal Encoding
- Countries (Nominal): One-Hot Encoding

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# Conclusions

- Pre-processing enhances the performance of data science models and understanding of the system being modeled.
- The selection of pre-processing techniques must be driven by the specific property of data that needs to be addressed.

## References and Further Reading

### Books:

- *Data Preprocessing in Data Mining* by Salvador García, Julián Luengo, Francisco Herrera

### Online Resource:

- Google Colab Notebook: Preprocessing Techniques in Python